

Geospatial & Drone Dataset Directory

Official catalog of drone orthomosaic imagery presets, national cadastral survey standards (SVAMITVA, DILRMP Bhu-Aadhaar ULPIN), and public computer-vision boundary training benchmarks.

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SAMPLE AERIAL IMAGERY

Ward 14 Shivajinagar Urban Compound Orthomosaic (Preset 1)

High-resolution nadir aerial survey photography covering urban residential parcel compounds, boundary walls, and access corridors.

Resolution: 4.8 cm/px GSD • Dimensions: 4096x3072 • CRS: EPSG:32643 (UTM 43N)

<https://images.unsplash.com/photo-1508873696983-2df5293cb325>

Ward 15 Aundh Agricultural Plot & Ridge Boundary (Preset 2)

Calibrated top-down aerial imagery displaying agricultural plot boundaries, hedgerows, and irrigation ditch ridges for SAM zero-shot delineation.

Resolution: 5.2 cm/px GSD • Dimensions: 3840x2880 • CRS: EPSG:32643 (UTM 43N)

<https://images.unsplash.com/photo-1524813686514-a57563d77d61>

Ward 16 Kothrud Metro Commercial Subdivision (Preset 3)

Nadir drone survey imagery of commercial land subdivisions, road setback alignments, and multi-tenant structural boundaries.

Resolution: 4.5 cm/px GSD • Dimensions: 4096x3072 • CRS: EPSG:32643 (UTM 43N)

<https://images.unsplash.com/photo-1500382017468-9049fed747ef>

CADASTRAL STANDARDS

SVAMITVA Drone Survey Scheme Portal

Survey of Villages and Mapping with Improvised Technology in Village Areas by Ministry of Panchayati Raj & Survey of India. Establishes the 5cm

Authority: Survey of India (Sol) • Ground Sample Distance: <= 5cm • Spatial Datum: WGS 84 / UTM

<https://svamitva.nic.in>

DILRMP & Bhu-Aadhaar (ULPIN) Standards

Digital India Land Records Modernization Programme by Department of Land Resources (DoLR), Ministry of Rural Development. Standardizes

ULPIN Formula: Lat/Long Centroid Encoding (14-Character Alphanumeric) • ISO 19152 LADM Aligned

<https://dilrmp.gov.in>

ISRO Bhuvan Indian Geo-Platform

National Remote Sensing Centre (NRSC) / ISRO spatial web portal providing Indian satellite imagery, topographic maps, village cadastral

Data Layers: Cartosat, LISS-IV, Bhuvan Thematic • Web Services: OGC WMS 1.3.0 / WFS 2.0.0

<https://bhuvan.nrsc.gov.in>

AI TRAINING DATASETS

OpenAerialMap (OAM) Global Drone Repository

Open-access, crowd-sourced repository of worldwide drone, UAV, and aerial orthomosaics with open GeoTIFF downloads and COG (Cloud

Format: GeoTIFF, COG, TileJSON • **License:** Creative Commons Attribution (CC-BY 4.0)

<https://openaerialmap.org>

SpaceNet Building Footprint & Parcel Extraction Datasets

Benchmark labeled geospatial datasets designed for automated computer-vision polygon boundary delineation, building footprint extraction, and

Data Volumes: 5+ Million Annotated Footprints • **Sensors:** WorldView-2, WorldView-3 (30cm/50cm)

<https://spacenet.ai>

Inria Aerial Image Labeling Benchmark

Gold-standard computer vision dataset for aerial semantic segmentation and boundary extraction across diverse urban and rural densities (0.3m

Coverage: 810 km² • **Cities:** Austin, Chicago, Kitsap, Vienna, Tyrol • **Labeled:** Building & Parcel Masks

<https://project.inria.fr/aerialimagelabeling/>

Meta AI Segment Anything (SAM) Dataset (SA-1B)

The dataset powering zero-shot boundary delineation and promptable segmentation (ViT-Huge / ViT-Large models) used for interactive boundary

Volume: 1.1 Billion Segmentation Masks • 11 Million High-Resolution Images • Zero-Shot Capable

<https://ai.meta.com/datasets/segment-anything/>

OPEN GEOSPATIAL REPOSITORIES

OpenStreetMap (OSM) Global Vector Geodata

Collaborative spatial database providing boundary polygons, parcel boundaries, land use classifications, and infrastructure corridors worldwide.

Format: GeoJSON, OSM PBF, Overpass API • **CRS:** EPSG:4326 (WGS 84)

<https://www.openstreetmap.org>

How to Use These Datasets with GeoHarmonize:

- 1. Upload any GeoTIFF, PNG, or JPEG from OpenAerialMap or your UAV flight using the [Upload Drone Image] tool.
- 2. The integrated SAM (Segment Anything) and Canny/Contour models will detect and bound parcel borders automatically.
- 3. Commit bounded vector borders into PostGIS with official 14-character Bhu-Aadhaar (ULPIN) identifiers.